What is static electricity?

Activity: Charge It

- Record your observations

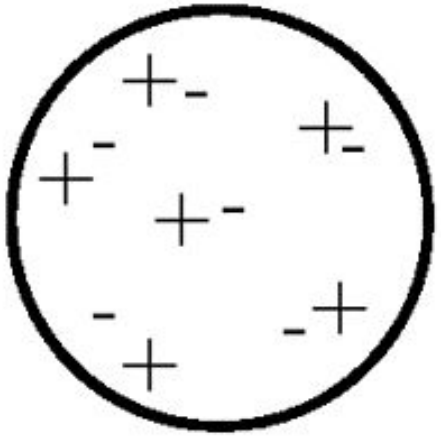
Electric Charges

Use the the image below to complete the chart in your handout…

- Objects become charged due to the movement of **electrons** , since it is the only particle found **outside** of the nucleus
- An object is **neutral** when there is an **equal** number of **protons** and **electrons**

6 positive charges and
6 negative charges

$6 + (-6) = 0$



There is zero net charge:
The object is neutral

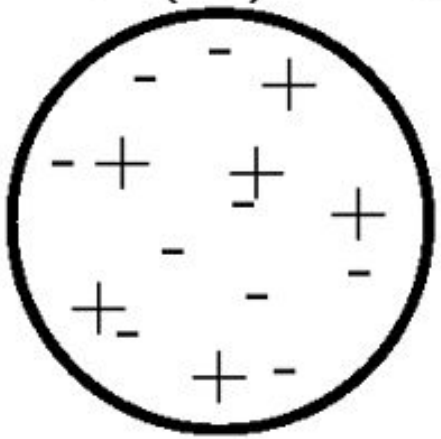
Electric Charges

Use the the image below to complete the chart in your handout…

- An object becomes **negatively** charged when it **gains** electrons

6 positive charges and
9 negative charges

6 + (-9) = -3



The net charge is -3

The object is negatively charged

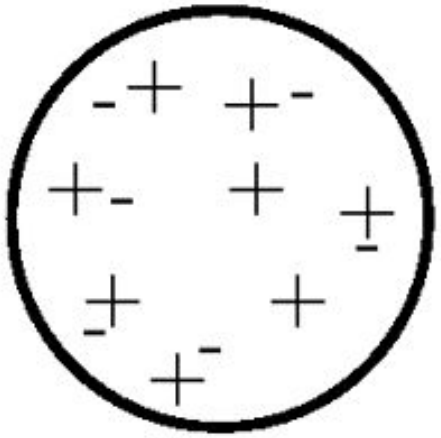
Electric Charges

Use the the image below to complete the chart in your handout…

- An object becomes **positively** charged when it **loses** electrons

8 positive charges and
6 negative charges

$8 + (-6) = 2$



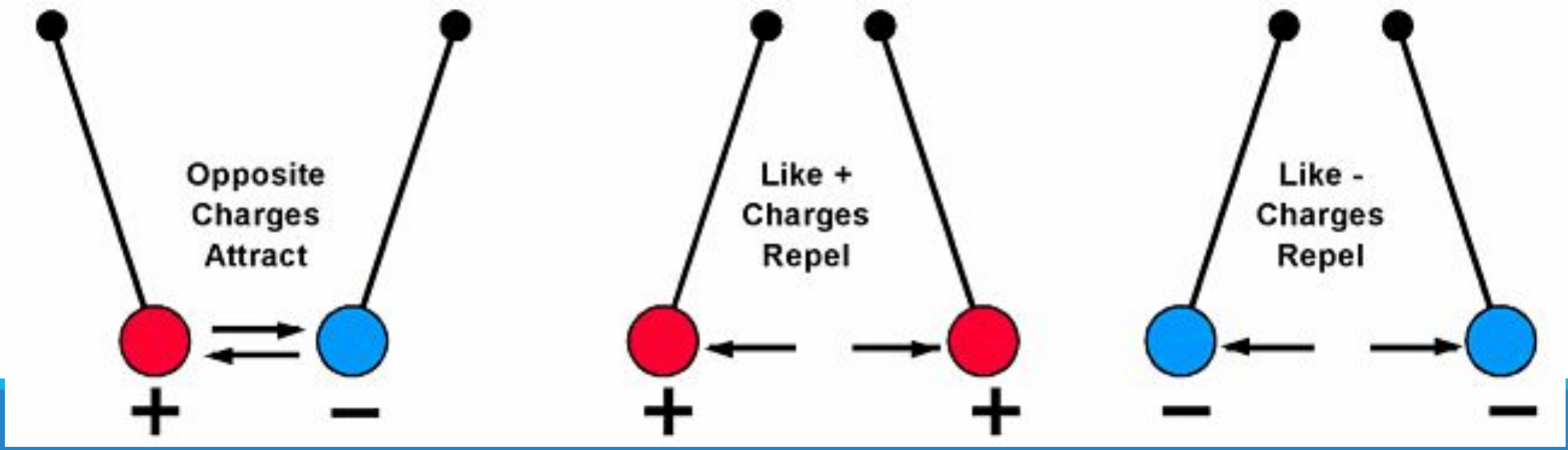
The net charge is +2

The object is positively charged

Law of Electric Charges

1) Objects that have **opposite** charges **attract**
2) Objects that have the **same** charges **repel** each other

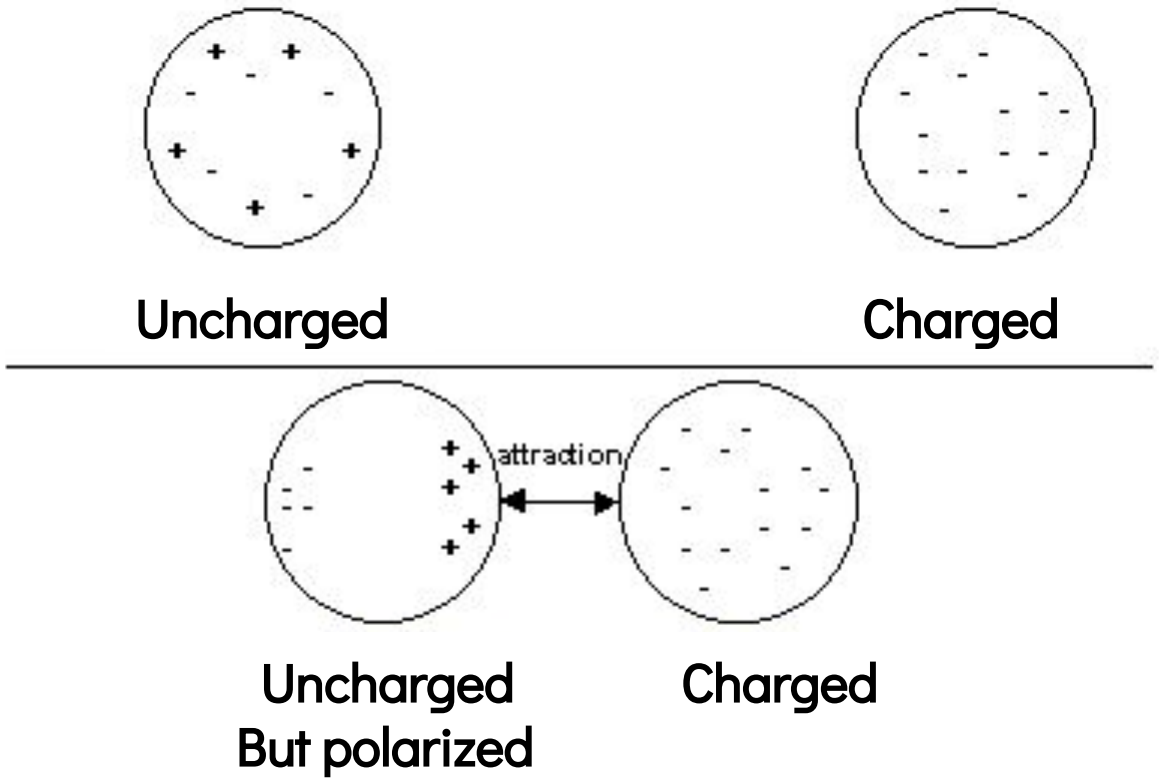
Use the the image below to complete the diagram in your handout…



Law of Electric Charges

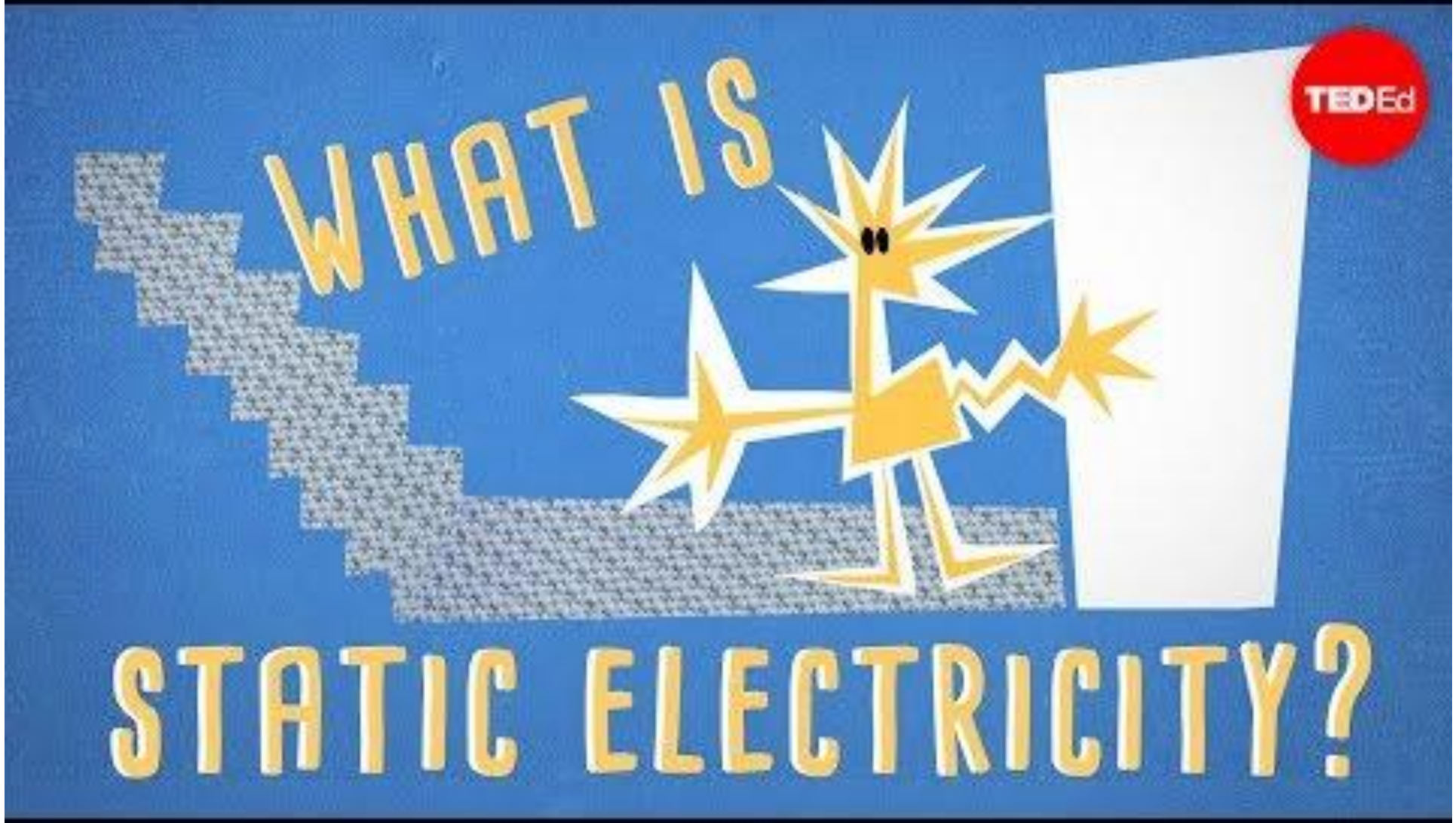
3) Charged objects **attract** neutral objects

Use the the image below to complete the diagram in your handout…



Static Electricity

- Static electricity is an **imbalance** of electric charge on the **surface** of an object
- "Static" means **not moving**
- An **electric discharge** is the sudden **movement** of static charge between two objects


WHAT IS
TEDEd
STATIC ELECTRICITY?

How do we charge objects?

Objects can be charged in three ways:

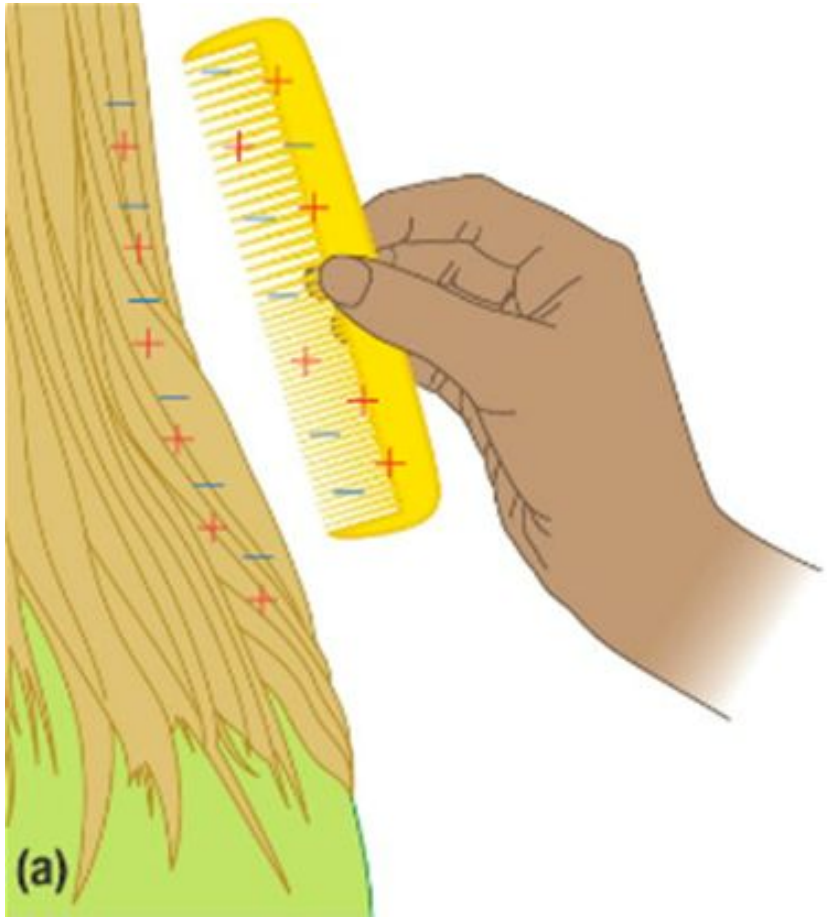
1. **Friction**
2. **Contact**
3. **Induction**

Charging by Friction

- Occurs when two **neutral** objects of **different** materials are **rubbed** together
- One of the objects will **lose** electrons and become **positive** , the other will **gain** electrons and become **negative**
- E.g. combing your hair

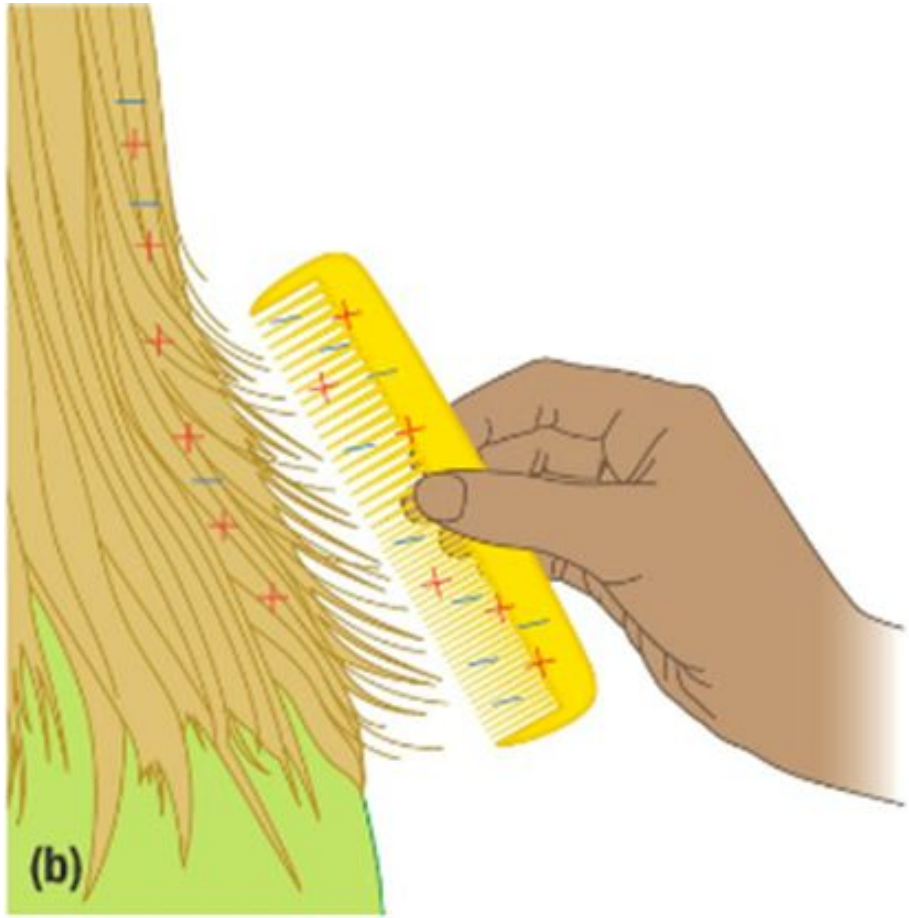
Charging by Friction

- The hair and the comb start out as electrically **neutral**



Charging by Friction

- Eventually the **comb** will attract **electrons** from your hair and become **negatively** charged and the **hair** will become **positively** charged



Predicting Electric Charge

- The **electrostatic series** helps you predict which static charge objects will acquire when rubbed together

Table 1 Electrostatic Series

Material	Charge tendency
human skin	+
rabbit fur	(weaker tendency to gain electrons)
acetate	
glass	
human hair	
nylon	
wool	
cat fur	
silk	
paper	
cotton	
wood	
amber	
rubber balloon	(stronger tendency to gain electrons)
vinyl	
polyester	
ebonite	–

Predicting Electric Charge

- The substance **higher** in the list will **lose** electrons (become **positive**)
- The substance **lower** in the list will **gain** electrons (become **negative**)

Table 1 Electrostatic Series

Material	Charge tendency
human skin	+
rabbit fur	(weaker tendency to gain electrons)
acetate	
glass	
human hair	
nylon	
wool	
cat fur	
silk	
paper	
cotton	
wood	
amber	
rubber balloon	(stronger tendency to gain electrons)
vinyl	
polyester	
ebonite	–

Example

You rub a rubber balloon on your head.

a) What charge does the balloon now have?

b) What charge does your hair now have?

Table 1 Electrostatic Series

Material	Charge tendency
human skin	+
rabbit fur	(weaker tendency to gain electrons)
acetate	
glass	
human hair	
nylon	
wool	
cat fur	
silk	
paper	
cotton	
wood	
amber	
rubber balloon	(stronger tendency to gain electrons)
vinyl	
polyester	
ebonite	−

Your turn

1) Finish the analysis questions from Activity: Charge It
2) Charged and Neutral Atoms/Charges on Objects
3) Static Charge Detective